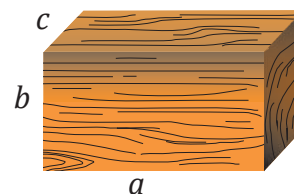


1 Sakkiz o'qli lokomotiv poyezdining massasi 160 tonna, har bir g'ildiragining relsga tegib turgan yuzasi 5 cm². Lokomotivning relsga beradigan bosimi qancha? Har bir o'qda 2 ta g'ildirak bor.

Berilgan:	Formula	Hisoblash
$m = 160 \text{ t} = 160000 \text{ kg}$ $S = 5 \text{ cm}^2 = 0,0005 \text{ m}^2$ $n = 16$ (g'ildiraklar soni) $g = 9,81 \frac{\text{N}}{\text{kg}}$	$p = \frac{F}{nS} = \frac{mg}{nS}$ $[p] = \frac{\text{kg} \cdot \frac{\text{N}}{\text{kg}}}{\text{m}^2} = \frac{\text{N}}{\text{m}^2} = \text{Pa}$	$p = \frac{160000 \cdot 9,81}{16 \cdot 0,0005} \text{ Pa} = 196200000 \text{ Pa}$ Javob: $p = 196,2 \text{ MPa}$.
Topish kerak: $p = ?$		

2 O'lchamlari 20 cm, 15 cm va 10 cm bo'lgan yog'och brusokning og'irligi 2,4 kg. Brusok turli holatlarda tayanchga qanday bosim beradi?



Berilgan:	Formula	Hisoblash
$m = 2,4 \text{ kg}$ $a = 20 \text{ cm} = 0,2 \text{ m}$ $b = 15 \text{ cm} = 0,15 \text{ m}$ $c = 10 \text{ cm} = 0,1 \text{ m}$ $g = 9,81 \frac{\text{N}}{\text{kg}}$	$S_1 = b \cdot c$ $S_2 = a \cdot b$ $S_3 = a \cdot c$ $p_1 = \frac{mg}{S_1}$ $p_2 = \frac{mg}{S_2}$ $p_3 = \frac{mg}{S_3}$	$S_1 = 0,15 \cdot 0,1 \text{ m}^2 = 0,015 \text{ m}^2$ $S_2 = 0,2 \cdot 0,15 \text{ m}^2 = 0,03 \text{ m}^2$ $S_3 = 0,2 \cdot 0,1 \text{ m}^2 = 0,02 \text{ m}^2$ $p_1 = \frac{2,4 \cdot 9,81}{0,015} \text{ Pa} = 1569,6 \text{ Pa}$ $p_2 = \frac{2,4 \cdot 9,81}{0,03} \text{ Pa} = 784,8 \text{ Pa}$ $p_3 = \frac{2,4 \cdot 9,81}{0,02} \text{ Pa} = 1177,2 \text{ Pa}$
Topish kerak: $p_1 = ?$ $p_2 = ?$ $p_3 = ?$	$[S] = a \cdot b = \text{m} \cdot \text{m} = \text{m}^2$ $[p] = \frac{\text{kg} \cdot \frac{\text{N}}{\text{kg}}}{\text{m}^2} = \frac{\text{N}}{\text{m}^2} = \text{Pa}$	Javoblar: $p_1 = 1569,6 \text{ Pa}$ $p_2 = 784,8 \text{ Pa}$ $p_3 = 1177,2 \text{ Pa}$

3 Uyning ayvoni to'rtta ustunga qurilgan. Har bir ustunning yuzasi 300 cm^2 . Ayvon tomiga yopilgan qurilish materiallarining massasi 720 kg bo'lsa, har bir ustun ayvon poliga qanday bosim beradi?

Berilgan:	Formula	Hisoblash
$S = 300 \text{ cm}^2$ $m = 720 \text{ kg}$ $n = 4$ $g = 9,81 \frac{\text{N}}{\text{kg}}$	$S_{um} = n S$ $p_{um} = \frac{m g}{S_{um}}$ $[S] = n \cdot S = \text{m}^2$	$S_{um} = 4 \cdot 0,03 \text{ m}^2 = 0,12 \text{ m}^2$ $p_{um} = \frac{720 \cdot 9,81}{0,12} \text{ Pa} = 58860 \text{ Pa}$ Har bir ustunning polga beradigan bosimi: $p = \frac{p_{um}}{4} = 14715 \text{ Pa}$
Topish kerak: $p = ?$	$[p] = \frac{\text{kg} \cdot \frac{\text{N}}{\text{kg}}}{\text{m}^2} = \frac{\text{N}}{\text{m}^2} = \text{Pa}$	Javob: $p = 14715 \text{ Pa}$.



13-mashq



1 Qanday sirt yuzasiga tik yo'nalishda 100 N kuch bilan ta'sir etib, 20 kPa bosim berish mumkin?

2 Belkurakning tig'li qismining eni 25 cm va qalinligi 1 mm ga teng bo'lib, odam belkurakka 50 N kuch bilan bossa, yerga qanday bosim beradi? Nima uchun belkurak tig'i o'tkir bo'lishi kerak?

3 Olti o'qli lokomotiv poyezdining massasi 120 tonna , har bir g'ildiragining relsga tegib turgan yuzasi 5 cm^2 . Uning relsga bosimi qancha? Har bir o'qda 2 ta g'ildirak bor.

4 Tikuv mashinasidagi igna 50 MPa bosimda ko'ylak tikadi. Agar igna uchining yuzasi $0,02 \text{ mm}^2$ bo'lsa, tikuv mashinasi qanday kuch bilan ignaga ta'sir ko'rsatadi?

5 Yengil avtomobilning massasi $1,8 \text{ tonna}$, umumiy ballonlarining yerga tegib turgan yuzasi 600 cm^2 . Samolyotning massasi 40 tonna , umumiy ballonlarining yerga tegib turgan yuzasi 2 m^2 . Ularning yerga beradigan bosimini hisoblang va taqqoslang.

6 Yengil avtomobil g'ildiraklarining yerga tegib turgan qismining yuzasi $0,5 \text{ m}^2$ ga teng. Avtomobilning massasi 2 tonna bo'lsa, har bir ballon yerga qanday bosim beradi?

7 Sportchi muz ustida konkida uchmoqda. Konkining uzunligi 30 cm , eni 2 mm . Sportchining massasi 48 kg bo'lsa, sportchining muzga beradigan bosimini toping.

8 Elektr dazmolning ishchi yuzasi 300 cm^2 ga teng bo'lsa, qanday bosim bilan sirtini bosadi? Dazmolning massasi $2,5 \text{ kg}$.

